Análisis NOS ID7_PM12

I. Identification

- **Study:** Cai H. et al., "Lower extremity muscle activation patterns in sarcopenic older adults during physical performance tests: implications for rehabilitation approaches." Scientific Reports (2025) 15:26932. DOI: 10.1038/s41598-025-10429-9.

- **Study design (context):** Cross-sectional, case (sarcopenic, n=8) vs age-matched control (healthy older adults, n=8) comparison using objective measures (BIA, grip strength, 6MWT/5STS) and surface EMG during 6MWT and 5STS. (See Methods: Participants; Sarcopenia assessment; Experiment protocol — pages 2–4.)

- **Main outcomes:** Spatiotemporal gait metrics, EMG-derived parameters (activation amplitudes, coactivation index, variability), task performance (6MWT, 5STS). (Results: Tables 1–6; pages 5–10.)

- **Data availability:** Data in manuscript and supplementary files; original data available on request from corresponding author (Data availability, page 13).

II. NOS assessment

Review 1: Using the Newcastle–Ottawa Scale for Case–Control Studies provided

1) Selection — Is the case definition adequate?

- **What it evaluates:** Whether cases are defined by valid, objective criteria or independent validation.

- **NOS criterion applied:** 1a (yes, with independent validation).

- **Evidence (quote + location):** "Participants were classified according to the AWGS 2019 consensus criteria… Muscle mass measured using InBody 970… grip strength measured by calibrated dynamometer… Sarcopenia was diagnosed when both low muscle mass and low muscle strength were present." (Methods: Sarcopenia assessment and classification; page 2)

- **Judgment R1:** Star awarded (yes — objective standardized diagnostic criteria and measurements).

- **Evidence R1 (quote/page):** see above (page 2).

2) Selection — Representativeness of the cases

- **What it evaluates:** Whether cases are a consecutive/representative series or likely selected.

- **NOS criterion applied:** 2a vs 2b.

- **Evidence:** "Sixteen older adults (8 with sarcopenia and 8 healthy controls) were recruited through university and community posters..." (Participants; page 2)

- **Judgment R1:** No star (potential for selection bias — volunteer/advertisement recruitment; small sample; not consecutive).

- **Evidence R1 (quote/page):** recruitment by posters (page 2).

3) Selection — Selection of Controls

- **What it evaluates:** Whether controls are community-based (preferred) or hospital-based.

- **NOS criterion applied:** 3a (community controls).

- **Evidence:** Controls were "age-matched healthy older adults" recruited through same recruitment channels (Participants; page 2) and compared as age-matched group throughout.

- **Judgment R1:** Star awarded (community/age-matched controls).

- **Evidence R1 (quote/page):** "Sixteen older adults (8 with sarcopenia and 8 healthy controls) were recruited through university and community posters..." (page 2).

4) Selection — Definition of Controls

- **What it evaluates:** Whether controls are confirmed free of the disease (endpoint).

- **NOS criterion applied:** 4a (no history of disease/endpoint).

- **Evidence:** Exclusion criteria removed conditions affecting functional performance; controls described as "healthy" and classified per AWGS thresholds (Methods; page 2).

- **Judgment R1:** Star awarded (controls defined and screened).

- **Evidence R1 (quote/page):** Exclusion criteria and AWGS classification (page 2).

Comparability

1) Comparability of cases and controls on design/analysis

- **What it evaluates:** Whether study controlled for the most important factor(s) (e.g., age).

- **NOS criterion applied:** 1a and possibly 1b (up to 2 stars).

- **Evidence:** Study used age-matched groups; no formal multivariable adjustment reported; authors note gender imbalance and no covariate analysis due to sample size (Discussion, limitations; page 12).

- **Judgment R1:** One star (controlled for age by matching; did not control/adjust for sex or other confounders).

- **Evidence R1 (quote/page):** "age-matched healthy controls" (Methods, page 2); "there was a gender imbalance… we did not perform a formal covariate analysis" (Discussion, page 12).

Exposure

1) Ascertainment of exposure

- **What it evaluates:** Whether exposure ascertainment is secure (records) or blinded interview, etc.

- **NOS criterion applied:** 1a or 1b (secure record / structured blinded interview).

- **Evidence:** Objective instrumentation: synchronized motion capture, force platforms, wireless surface EMG; MVC normalization; standardized protocols (Experiment protocol; EMG methods; pages 3–4).

- **Judgment R1:** Star awarded (secure, objective measurement of EMG/biomechanics).

- **Evidence R1 (quote/page):** EMG and instrumentation details (pages 3–4).

2) Same method of ascertainment for cases and controls

- **What it evaluates:** Whether exposure measured identically in both groups.

- **NOS criterion applied:** 2a (yes).

- **Evidence:** All participants underwent same standardized protocol, same examiner, same trials (Experiment protocol; page 2–4).

- **Judgment R1:** Star awarded (identical measurement procedures).

- **Evidence R1 (quote/page):** "All measurements were carried out by the same examiner… Each participant performed five successful 6MWT trials and three 5STS trials…" (pages 2–3).

3) Non-response rate

- **What it evaluates:** Whether non-response/refusal rates are reported and comparable.

- **NOS criterion applied:** 3a/3b/3c.

- **Evidence:** No reporting of recruitment/enrollment refusal rates or detailed non-response description in manuscript or supplement; small sample and recruitment by posters.

- **Judgment R1:** No star (non-response not described).

- **Evidence R1 (quote/page):** absence of recruitment/non-response data; recruitment by posters only (page 2).

Overall NOS summary

- **Stars awarded:** Selection = 3/4 (items 1,3,4), Comparability = 1/2, Exposure = 2/3. Total = 6/9 stars.

- **General NOS judgment (standard options):** Fair quality (moderate risk of bias).

- **Justification:** Objective case definition and robust, identical exposure measurement methods support internal validity; however, volunteer recruitment, very small sample size (n=16), lack of adjustment for sex and other confounders, and absent non-response reporting limit representativeness and increase risk of selection/confounding bias. (Evidence: Methods, Results, Discussion; pages 2–13.)

III. NOS assessment

Review 2 (R2) (independent verification)

1) Case definition adequate

- Evidence: AWGS criteria, BIA, grip strength, objective thresholds (page 2).

- Judgment R2: Star awarded.

- Evidence R2 (quote/page): same as R1 (page 2).

2) Representativeness of cases

- Evidence: recruitment via posters; authors state pilot/exploratory, small sample and gender imbalance (pages 2, 12).

- Judgment R2: No star (confirmed potential selection bias).

- Evidence R2 (quote/page): recruitment method and limitations (pages 2, 12).

3) Selection of controls

- Re-check: Age-matched community controls recruited similarly (page 2).

- Judgment R2: Star awarded.

- Evidence R2 (quote/page): participants recruited through posters; described as healthy controls (page 2).

4) Definition of controls

- Re-check: Exclusion criteria and screening described (page 2).

- Judgment R2: Star awarded.

- Evidence R2 (quote/page): exclusion criteria and classification (page 2).

Comparability

- Re-check: age-matching present; no multivariable control for sex or other variables; authors note imbalance (page 12).

- Judgment R2: One star (consistent with R1).

- Evidence R2 (quote/page): age-matching statement and limitation text (pages 2,12).

Exposure ascertainment

- Re-check: rigorous instrumentation, MVC normalization, SENIAM guidelines (pages 3–4).

- Judgment R2: Star awarded.

- Evidence R2 (quote/page): EMG instrumentation and processing (pages 3–4).

Same method for cases/controls

- Re-check: identical testing procedure and examiner (pages 2–3).

- Judgment R2: Star awarded.

- Evidence R2 (quote/page): standardized demonstrations, same examiner (page 2).

Non-response rate

- Re-check: no mention of refusals or response rates anywhere; sample small (pages 2,12,13).

- Judgment R2: No star.

- Evidence R2 (quote/page): absence of reporting; authors do not report recruitment numbers screened vs enrolled.

General consensus proposal and justification

Consensus judgment (final): 6/9 stars — Fair quality (moderate risk of bias).

Consensus comment (brief):

Strengths: Objective, standardized case definition and high-quality, identical exposure measurement (EMG and biomechanical instrumentation).

Limitations: A non-representative volunteer recruitment (posters), very small sample size (n=16) with acknowledged gender imbalance and no multivariable confounder control, and absent reporting of non-response/enrollment flow. These limitations increase selection and confounding risk despite rigorous measurement methods.

(Evidence locations: Methods: Participants & Sarcopenia classification, Experiment protocol, EMG processing — pages 2–4; Results/Tables — pages 5–10; Discussion limitations — page 12; Data availability — page 13.)

Final concise summary

Final NOS score: 6/9 stars — Overall quality: Fair (moderate risk of bias) due to robust objective measures and case definition but limited by volunteer recruitment, very small sample, gender imbalance, and lack of adjustment/reporting for non-response and other confounders (evidence: Methods pages 2–4; Results pages 5–10; Discussion pages 12–13).

IV. XLSX-summary

(Columns: Study ID	*Domain*	*Item description*	*Judgment R1*	*Evidence R1 (quote/page)*	*Judgment R2*	*Evidence R2*	*Consensus judgment*	*consensus comment)*
Study ID	Domain	Item description	Judgment R1	Evidence R1 (quote/page)	Judgment R2	Evidence R2	Consensus judgment	consensus comment
Cai2025	Selection (Case definition)	Is the case definition adequate? (objective, validated diagnosis)	Star	"Participants were classified according to the AWGS 2019 consensus criteria... Muscle mass measured using InBody 970... grip strength measured by calibrated dynamometer... Sarcopenia was diagnosed when both low muscle mass and low muscle strength were present." (Methods, page 2)	Star	same quote (Methods, page 2)	Star	Objective AWGS-based criteria with BIA and dynamometer used — qualifies for NOS star.
Cai2025	Selection (Representativeness)	Representativeness of the cases (consecutive/representative series)	No star	"recruited through university and community posters" (Participa	No star	same (Participants, page 2); plus limitation note (Discus	No star	Recruitment via posters (volunteer sample), small pilot sample and gender imbalance

				nts, page 2)		sion, page 12)		— potential selection bias.
Cai2025	Selection (Controls selection)	Selection of controls (community vs hospital)	Star	"Sixteen older adults (8 with sarcopenia and 8 healthy controls) were recruited through university and community posters..." (Participants, page 2)	Star	same (Participants, page 2)	Star	Controls were community age-matched participants recruited similarly — acceptable control selection.
Cai2025	Selection (Definition of controls)	Definition of controls (no history of disease/endpoint)	Star	Exclusion criteria removed conditions affecting functional performance; controls described as healthy (Methods, page 2)	Star	same (Methods, page 2)	Star	Controls screened via exclusion criteria and classified; defined as free from target condition.
Cai2025	Comparability	Comparability of cases and controls (controls for most important factor(s))	1 star	Age-matched groups; no covariate adjustment; "gender imbalance... we did not perform a formal covariate analysis" (Methods & Discussion, pages 2,12)	1 star	same (Methods & Discussion, pages 2,12)	1 star	Age matching performed (major confounder controlled) but no adjustment for sex/other confounders due to small sample — only one star.
Cai2025	Exposure (Ascertainment)	Ascertainment of exposure (secure records/objective measurement)	Star	Detailed instrumentation: synchronized	Star	same (pages 3–4)	Star	Objective, standardized EMG and biomechanical

				motion capture, force plates, Trigno Avanti EMG, MVC normalization, SENIAM guidelines (Experiment protocol & EMG methods, pages 3–4)				measurements — secure ascertainment.
Cai2025	Exposure (Same method)	Same method of ascertainment for cases and controls	Star	"All measurements were carried out by the same examiner... Each participant performed five successful 6MWT trials and three 5STS trials..." (Experiment protocol, pages 2–3)	Star	same (pages 2–3)	Star	Identical standardized protocol applied to both groups.
Cai2025	Exposure (Non-response rate)	Non-response / participation rate described and comparable	No star	No reporting of refusals, screened vs enrolled numbers; recruitment by posters only (no flow data) (Participants & Data availabilit	No star	same (pages 2,13)	No star	Non-response/enrollment flow absent; cannot assess potential participation bias.

	y, pages 2,13)

V. Final concise summary

Final NOS score: 6/9 stars — Overall quality: Fair (moderate risk of bias) due to robust objective measures and case definition but limited by volunteer recruitment, very small sample, gender imbalance, and lack of adjustment/reporting for non-response and other confounders (evidence: Methods pages 2–4; Results pages 5–10; Discussion pages 12–13).